Legend
- Line 151
- Line 152
- Line 153
- Line 154
- Line 155
- Line 156
- Line 157
- Line 158
- Line 159
- Line 160
- Line 161
- Line 162
- Line 163
- Line 164
- Line 165

101-E-102A/B/C/D
CRUDE/CRUDE COLUMN OVERHEAD EXCHANGER

(TYPE-C, NOTE-1)
S/S DT - 175° C
DP - 36.7 kg/cm²g
T/S DT - 161.5° C
DP - 28.2 kg/cm²g
DESIGN DUTY - 26.74 MMkcal/hr

4" -FL-101-1204-A11A

3"-FL-101-1205-A11A

4"x3"

3"-WP-101-4811-B19A-IH

FROM 101-E-130A/B
WASH WATER
101-1148

TW 1212 | PI 1205 | TE 1223 | TT 1223 | FF | TI 1223

NOTE-3,5

NO POCKETS

48"-P-101-2701-A19A-IH

FROM 101-C-101
CR COL OVHD VAP
101-1127

MAKE CONNECTION AT MID POINT OF HEADER FROM TOP

MIN.

36"-P-101-1210-A19A-IH

NOTE-3,5

36"x40" | 40"x36" | 48"x40" | 48"x40"

NOTE-3,5

36"-P-101-1211-A19A-IH

MIN.

TI 1205 | TT 1205 | TE 1205 | PI 1201 | OP | TW 1210

24"-P-101-1209-B1A-IT
TO 101-E-103A/B & 105A/B
CRUDE
101-1113

NOTE-6

14"x20" | 20"x24"

FI 1201 A | FI 1201 B | FT 1201 | FE-1201 | FACING | SLOPE 1:500 | 26"-P-101-1212-A19A-IH | TI 1201 | TT 1201 | TE 1201 | 12"-P-101-1205-B1A-IT | 2"-SV-101-1201-B1A/T | 1.5E2 | 36.7 | PSV 1201 | 900# | 2"X1-1/2" | 3/4" | LO FB | 2"-WP-101-1201-B19A-IH | NC | SP 036 | NOTE-4 | B19A | A19A | A11A | IH | 2"-P-101-1220-B1A-IT | MIN. | 26"x24" | 300# | T1 | T2 | S1 | S2 | A

12"-P-101-1206-B1A-IT | TI 1202 | TT 1202 | TE 1202 | 2"-SV-101-1202-B1A/T | 1.5E2 | 36.7 | PSV 1202 | 900# | 2"X1-1/2" | 3/4" | LO FB | 26"-P-101-1215-A19A-IH | SLOPE 1:500 | SP 110 | NOTE-4 | A19A | B19A | FI 1202 A | FI 1202 B | FT 1202 | FE-1202 | FACING | 2"-WP-101-1202-B19A-IH | 3"-FL-101-1202-A11A | 2" | CP 1210 | CC | SP 010 | 2"-P-101-1221-B1A-IT | B

3"-FL-101-1203-A11A | FI 1203 A | FI 1203 B | FT 1203 | FE-1203 | FACING | SLOPE 1:500 | 26"-P-101-1214-A19A-IH | TI 1203 | TT 1203 | TE 1203 | 12"-P-101-1207-B1A-IT | 2"-SV-101-1203-B1A/T | 1.5E2 | 36.7 | PSV 1203 | 900# | 2"X1-1/2" | 3/4" | LO FB | 2"-WP-101-1203-B19A-IH | SP 111 | NOTE-4 | B19A | A19A | A11A | IH | 2"-P-101-1222-B1A-IT | C

12"-P-101-1208-B1A-IT | TI 1204 | TT 1204 | TE 1204 | 2"-SV-101-1204-B1A/T | 1.5E2 | 36.7 | PSV 1204 | 900# | 2"X1-1/2" | 3/4" | LO FB | 26"-P-101-1215-A19A-IH | SLOPE 1:500 | 2"-WP-101-1204-B19A-IH | FI 1204 A | FI 1204 B | FT 1204 | FE-1204 | FACING | 3"-FL-101-1204-A11A | SP 112 | NOTE-4 | A19A | B19A | NOTE-8 | 2" | CP 1212 | CC | SP 016 | 2"-P-101-1223-B1A-IT | D

14"-P-101-1208-B1A-IT | TT 1206 | TE 1206 | TI 1206 | PI 1209 | NOTE-8 (TYP.) | SP 008 | CP 1205 | 12"-P-101-1201-B1A-IT

12"-P-101-1202-B1A-IT | TE 1207 | TT 1207 | TI 1207 | PI 1206 | SP 012 | CP 1206

TI 1208 | TT 1208 | TE 1208 | PI 1212 | SP 014 | CP 1207 | 12"-P-101-1203-B1A-IT

12"-P-101-1204-B1A-IT | TE 1209 | TT 1209 | TI 1209 | PI 1215 | SP 018 | CP 1208

101-E-102A/B/C/D

FROM 101-E-101A/B
CRUDE
101-1111

24"-P-101-1113-B1A-IT

NOTE-6

16"x12" | 24"x16"

TWO PHASE FLOW | NOTE-5,10 | NO POCKETS | 24"-P-101-1219-A19A-IS | TO 101-A-101 D | CR COL OVHD | 101-1128

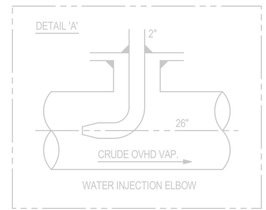
TWO PHASE FLOW | NOTE-5,10 | NO POCKETS | 24"-P-101-1218-A19A-IS | TO 101-A-101 C | CR COL OVHD | 101-1128

TWO PHASE FLOW | NOTE-5,10 | NO POCKETS | 24"-P-101-1217-A19A-IS | TO 101-A-101 B | CR COL OVHD | 101-1128

TWO PHASE FLOW | NOTE-5,10 | NO POCKETS | 24"-P-101-1216-A19A-IS | TO 101-A-101 A | CR COL OVHD | 101-1128

NOTES: -

1. FOR LEGEND SYMBOL AND GENERAL DETAILS, REFER P&ID NO. B068-79-41-00-1191/1192. (TPL DWG. NO. B224-CDU-TPL-000-1191/1192-SHT01/1192-SHT02).
   - FOR EXCHANGER DRAIN DETAILS REFER P&ID NO. B224-CDU-TPL-101-1246-SHT01 & B224-CDU-TPL-101-1246-SHT02.
   - FOR AIR COOLER DRAIN CONNECTION DETAILS REFER P&ID NO. B224-CDU-TPL-101-1247.
   - FOR PUMP DRAIN/VENT REFER P&ID NO.B224-CDU-TPL-101-1245-SHT01 & B224-CDU-TPL-101-1245-SHT02.
   - FOR FILTER DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1248.
   - FOR PUMP SEAL PLAN DETAIL REFER P&ID NO. B224-CDU-TPL-101-1184-SHT01, B224-CDU-TPL-101-1184-SHT02 & B224-CDU-TPL-101-1184-SHT03.
   - FOR CONTROL VALVE DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1243-SHT01 & B224-CDU-TPL-101-1243-SHT02.
   - FOR LEVEL INSTRUMENT DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1244-SHT01 & B224-CDU-TPL-101-1244-SHT02.
   - ALL INSTRUMENT TAG NUMBERS TO HAVE THE UNIT DESIGNATION PREFIX 101.
2. ALL DCS OPEN AND CLOSED LOOP INSTRUMENTS SHOWN IN P&ID WITH 'FF' ARE FOUNDATION FIELD BUS DEVICES. FOR INSTRUMENTS FOR WHICH FIELD BUS DEVICES ARE NOT SHOWN/AVAILABLE, SMART TRANSMITTERS/SMART POSITIONERS SHALL BE USED
3. SYMMETRICAL PIPING.
4. WASH WATER TO BE INJECTED WITH REVERSE FLOW, CONE SPRAY INJECTION NOZZLE, AS PER DETAIL SHOWN BELOW.



THE INJECTION NOZZLE TO BE INSERTED IN HORIZONTAL PORTION FOR PROPER MIXING

5. MAXIMUM ALLOWABLE PRESSURE DROP IN THE LINES IN THE CRUDE COLUMN OVHD CIRCUIT (I.E. FROM 101-C-101 TO 101-V-101 VIA 101-E-102A/B/C/D AND 101-A-101) IS 0.1 KG/CM2. ALL LINE SIZES IN THE CIRCUIT ARE CONFIRMED BASED ON FINAL ISOMETRICS.
6. CRUDE INLET AND OUTLET LINES TO BE ORIENTED SUCH THAT THE INLET AND OUTLET ARE DIAGONALLY OPPOSITE TO EACH OTHER AS SHOWN ON P&ID.
7. CORROSION MONITORING SYSTEM (CMS) SHALL BE DONE WITH CORROSION PROBE & CORROSION COUPON.
8. A MIN. OF 1 TO 1.5 M DISTANCE SHOULD BE KEPT BETWEEN CP AND CC. FOR MAINTENANCE PURPOSE.
9. CHEMICAL INJECTION SHALL BE DOSED THROUGH INJECTION TUBE WITH T-TYPE ACCESS FITTINGS.
10. LINE TO BE PROPERLY SUPPORTED.

No.	DATE	REVISIONS	DRN	BY	CKD	APPD
4	10.03.22	ISSUED FOR CONSTRUCTION	SC	MW	YD	PV
3	30.09.21	ISSUED FOR CONSTRUCTION	SC	MW	YD	PV
2	02.09.20	ISSUED FOR ENGG/ DESIGN WITH HAZOP RECOMMENDATION	RMB	KG	RS	SG
1	11.08.20	ISSUED FOR ENGINEERING/DESIGN	JO	KG	RS	SG
0	13.03.20	ISSUED FOR REVIEW	LC	KG	RS	SG

No.	DATE	REVISIONS	DRN	BY	CKD	APPD
1	14.12.18	REVISED AS MARKED	PP	UA/AT	GN	AM
0	15.11.18	ISSUED FOR PACKAGE	B.K.L	UA/AT	GN	AM

DETAILED ENGINEERING CONTRACTOR: FLUOR — FLUOR DANIEL INTERNATIONAL (MALAYSIA) SDN BHD

LSTK CONTRACTOR: TATA PROJECTS LIMITED, MUMBAI

PMC: ENGINEERS INDIA LIMITED (A Govt. of India Undertaking)

OWNER: एच पी सी एल राजस्थान रिफाइनरी लिमिटेड
HPCL RAJASTHAN REFINERY LTD
RAJASTHAN REFINERY PROJECT

पाइपिंग एण्ड इंस्ट्रूमैन्टेशन डायाग्राम
PIPING AND INSTRUMENTATION DIAGRAM
CRUDE/VACUUM DISTILLATION UNIT
CRUDE PREHEAT TRAIN-1B

SCALE	JOB NO.	DEPT.	SECTN.	UNIT	DWG. No.	REV.
	B068	79	41	101	1112	1

TPL DRAWING NO. B224-CDU-TPL-101-1112 REV 4

DOCUMENT REVIEW DETAILS
1 No Comments. Doc. B224-CDU-TPL-101-1112 Rev. 4
2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
3 Document does not conform to basic requirements marked. Resubmit for review.
R Received for Records. V Void.
Signature with Date: 24-AUG-2022
Name: Reviewer: UMER AFZAL (C889)
Page No.: 1 of 2

NOTICE: THIS DRAWING AND THE DESIGN IT COVERS ARE THE PROPERTY OF ENGINEERS INDIA LIMITED. THEY ARE MERELY LOANED AND ON THE BORROWER'S EXPRESS AGREEMENT THAT THEY WILL NOT BE USED, COPIED, LOANED, EXHIBITED, NOR USED EXCEPT IN THE LIMITED WAY AND PRIVATE USE PERMITTED BY ANY WRITTEN CONSENT GIVEN BY LENDER TO THE BORROWER.